



Department of Computer Science
Faculty of Computing
UNIVERSITI TEKNOLOGI MALAYSIA

SUBJECT NAME: COMPUTER ORGANIZATION AND ARCHITECTURE

SUBJECT CODE: SECR1033 / SCSR1033

SEMESTER: 2 – 2025/2026

LAB TITLE: Programming 1: Assembly Language Fundamentals

INSTRUCTION: Student is required to have instructor's signature before proceed from each lab work.

STUDENT INFO :

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COMMENTS:

Lab # 1

Simple program to familiarize with Program Code, Rebuild & Start Without Debugging

Execute the programs below:

PART 1:

i. Part A: Adding and Subtracting Integers

```
TITLE Add and Subtract (AddSub.asm)

; This program adds and subtracts 32-bit integers
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc

.data
TOTAL dword 0 ; a variable named TOTAL (declared as DWORD)

.code

main PROC

    mov eax, 123400h ; Set EAX with the value of 123400h
    add eax, 567800h ; Add the content of EAX with 567800h
    sub eax, 77700h ; Subtract content of EAX with 77700h
    mov TOTAL, eax ; Store content of EAX to TOTAL

    call DumpRegs

    exit
main ENDP

END main
```

Experimental Results:

a) Rebuild & Start Without Debugging



Completed Fully Partially : Checked by: _____

Screenshot Result:

```
EAX=00613500  EBX=00D08000  ECX=009710AA  EDX=009710AA
ESI=009710AA  EDI=009710AA  EBP=00BBF830  ESP=00BBF824
EIP=00973679  EFL=00000206  CF=0  SF=0  ZF=0  OF=0  AF=0  PF=1

C:\Users\User\OneDrive\Desktop\assembly coding\Project1\Debug\templateASM.exe (process 6840) exited with code 0.
Press any key to close this window . . .
```

ii. Part B: Adding Variables

```
TITLE Add and Subtract, Version 2 (AddSub2.asm)

; This program adds and subtracts 32-bit unsigned
; integers and stores the sum in a variable.
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc
.data
val1 DWORD 10000h
val2 DWORD 40000h
val3 DWORD 20000h
finalVal DWORD ?
.code

main PROC
    mov eax, val1      ; start with 10000h
    add eax, val2      ; add 40000h
    sub eax, val3      ; subtract 20000h
    mov finalVal, eax ; store the result (30000h)

    call DumpRegs     ; display the registers

    exit

main ENDP
END main
```

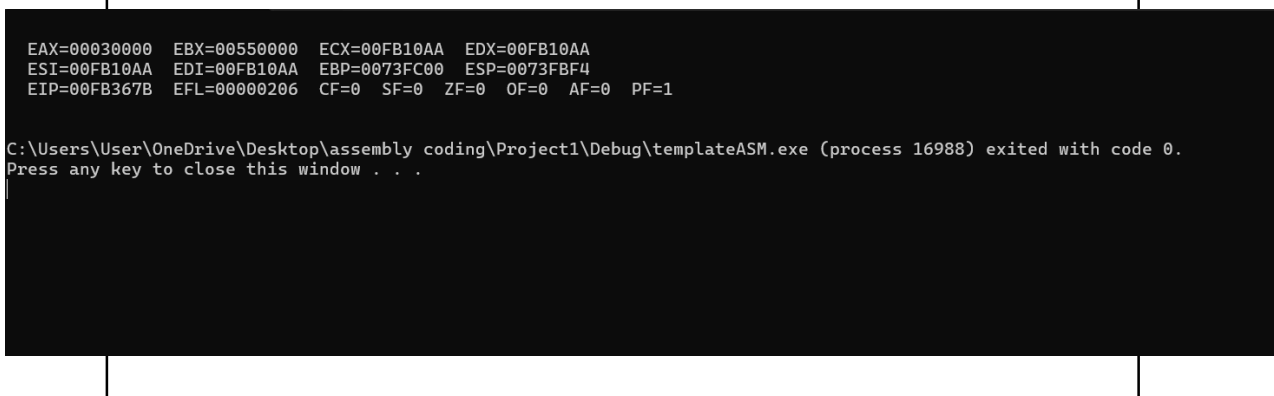
Experimental Results:

a) Rebuild & Start Without Debugging



Completed *Fully* *Partially* : Checked by: _____

Screenshot Result:



iii. Part C: Add and Subtract 8 and 16-Bit Version

```
TITLE Add and Subtract, Version 3
; This program adds and subtracts 8 and 16 bit
; unsigned integers and stores the sum in a variable.
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc

.data
valw1 WORD 1000h
valw2 WORD 4000h
valw3 WORD 2000h
finalValw WORD ?

valb1 BYTE 10h
valb2 BYTE 40h
valb3 BYTE 20h
finalValb BYTE ?

.code
main PROC
mov ax,valw1; start with 10000h
add ax,valw2      ; add 40000h
sub ax,valw3      ; subtract 20000h
mov finalValw,ax ; store the result (30000h)
call DumpRegs    ; display the registers

mov ah,valb1; start with 10000h
add ah,valb2      ; add 40000h
sub ah,valb3      ; subtract 20000h
mov finalValb,ah ; store the result (30000h)
call DumpRegs    ; display the registers

exit
main ENDP
END main
```

Experimental Results:

a) Rebuild & Start Without Debugging



Completed Fully Partially : Checked by: _____

Screenshot Result:

```
EAX=005C3000 EBX=0071E000 ECX=00B510AA EDX=00B510AA
ESI=00B510AA EDI=00B510AA EBP=005CFBE4 ESP=005CFBD8
EIP=00B5367F EFL=00000206 CF=0 SF=0 ZF=0 OF=0 AF=0 PF=1

EAX=005C3000 EBX=0071E000 ECX=00B510AA EDX=00B510AA
ESI=00B510AA EDI=00B510AA EBP=005CFBE4 ESP=005CFBD8
EIP=00B5369C EFL=00000206 CF=0 SF=0 ZF=0 OF=0 AF=0 PF=1

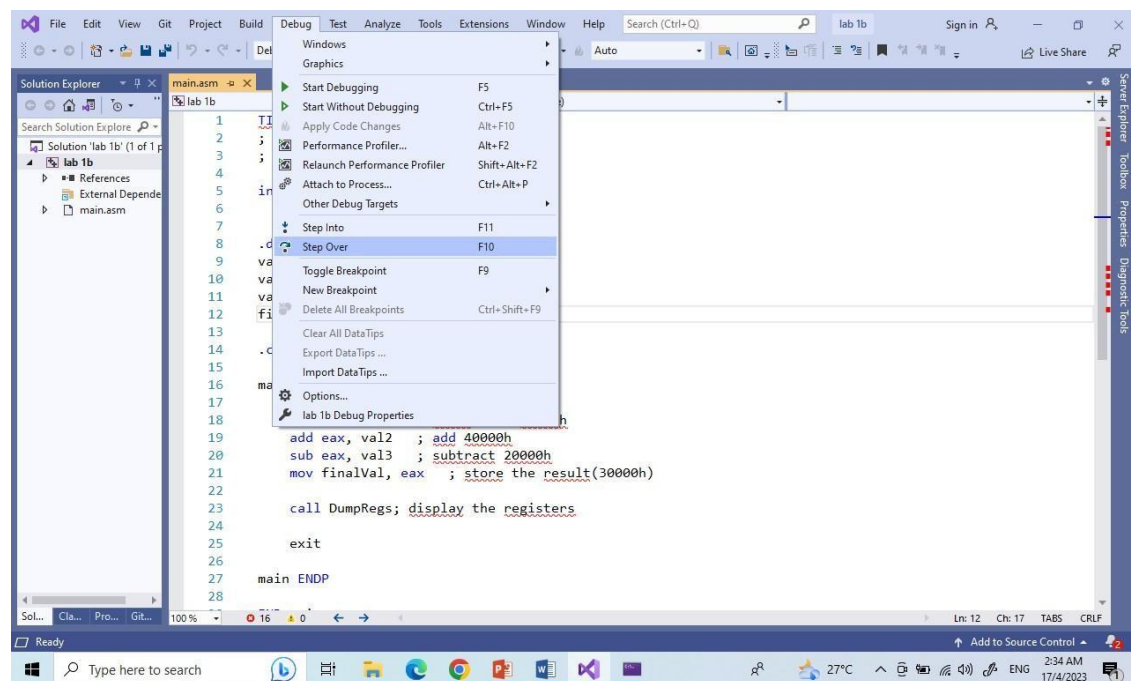
C:\Users\User\OneDrive\Desktop\assembly coding\Project1\Debug\templateASM.exe (process 14116) exited with code 0.
Press any key to close this window . . .
```

Detail Debugging Process

PART 2:

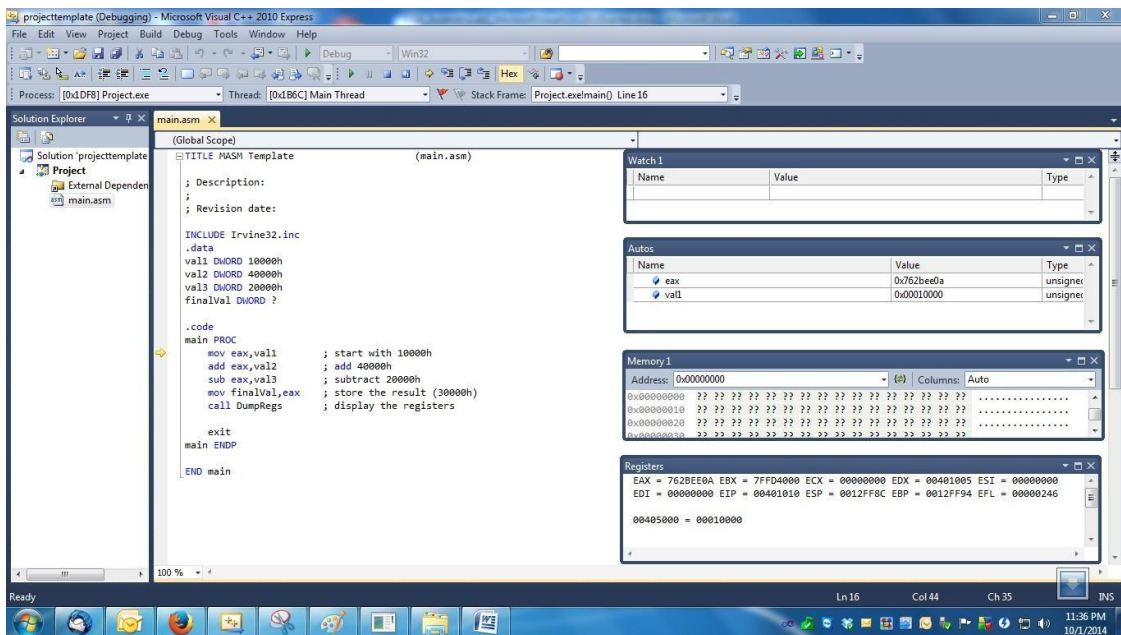
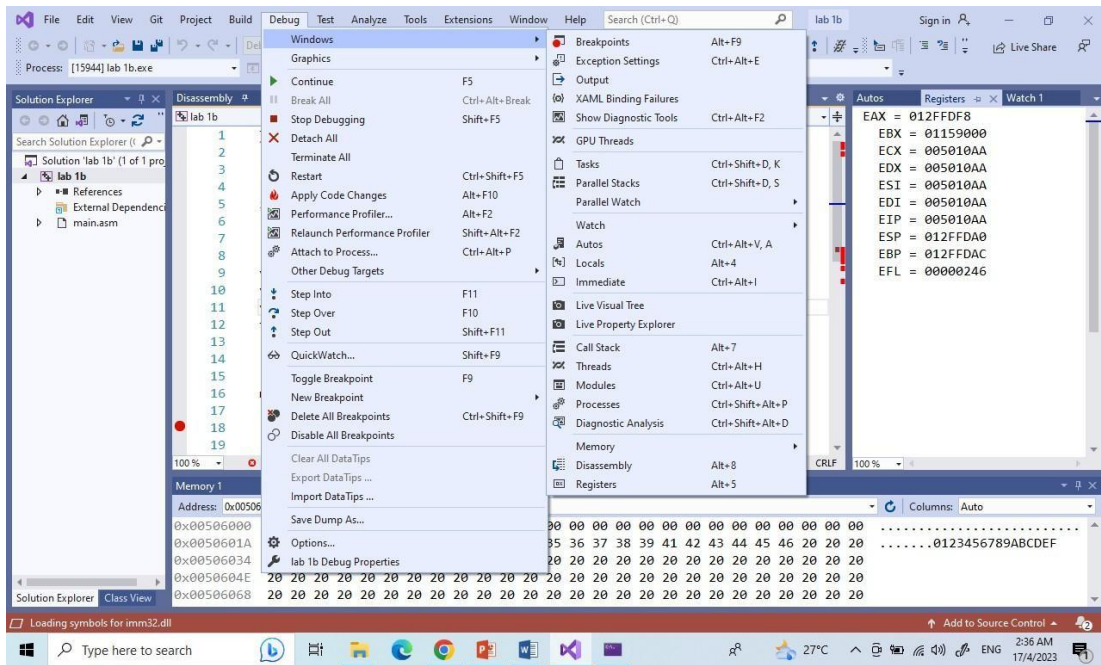
Run Debugging Process for Part B, Part C and Capture Video

1. Press F10 for step by step debugging.

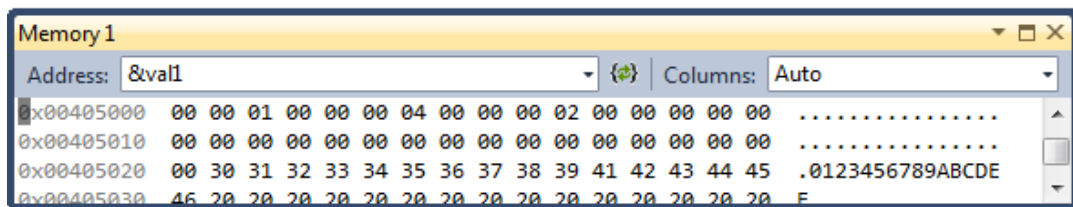


3. Open 5 windows:

- a. Watch
- b. Autos
- c. Memory
- d. Registers
- e. Disassembly



4. Default memory data segment value is at first variable: &val1 or 0x00405000 (depend on memory used).



5. Add variables val1, val2, val3 and finalval to Watch

Watch 1		
Name	Value	Type
val1	0x00010000	unsigned long
val2	0x00040000	unsigned long
val3	0x00020000	unsigned long
finalVal	0x00000000	unsigned long

6. F10 to trace/step the assembly program line by line.

7. Please debug by looking at value changes at Watch, Memory, Autos and Registers windows.

8. Disassembly window can show the following optional information:

- Memory address where each instruction is located. For native applications, it is the actual memory address. For Visual Basic or C#, it's an offset from the beginning of the function.
- Source code from which the assembly code derives.
- Code bytes, that is, the byte representations of the actual machine or MSIL instructions.
- Symbol names for the memory addresses.
- Line numbers corresponding to the source code.

What is Byte Code (or Machine Code) for the following assembly instructions:

```
mov ax, valw1 ; start with 10000h
add ax, valw2 ; add 40000h
sub ax, valw3 ; subtract 20000h
mov finalValw, ax ; store the result (30000h)
```

Byte Code:

```
mov ax, valw1 ; start with 10000h
005B3660 mov ax, word ptr [valw1 (05B6000h)]
add ax, valw2 ; add 40000h
005B3666 add ax, word ptr [valw2 (05B6002h)]
sub ax, valw3 ; subtract 20000h
005B366D sub ax, word ptr [valw3 (05B6004h)]
mov finalValw, ax ; store the result (30000h)
005B3674 mov word ptr [finalValw (05B6006h)], ax
call DumpRegs ; display the registers
005B367A call _DumpRegs@0 (05B10E1h)
```

Experimental Results:

a) Rebuild & Start Debugging

CamStudio:



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VIDEO LINK :
[video](#)